Can superparamagnetic iron-oxide nanoparticles be seen in CT?

Andreas Maier[a], Stefan Lyer[b], Lukas Heinen[b], and Rainer Tietze[b]

[a]Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany
[b]SEON – Section of Experimental Oncology and Nanomedicine, Department of Otorhinolaryngology, Universitätsklinikum Erlangen / FAU, Germany

ABSTRACT

Superparamagnetic iron-oxide nanoparticles (SPIONs) are used for magnetic drug targeting, hyperthermia, and cell tracking, but a simple question is often left open: can the deposited iron be seen in a computed tomography (CT) scan? We answer this by simulation, starting from a real magnetite SPION formulation and building a delivered-mass dose model in which a fixed particle mass is spread through an 8 cm^3 tumor. At realistic loading this leaves only a fraction of a milligram of iron per millilitre, about ten times below iodine CT enhancement, and because iron has no usable K-edge in the diagnostic range its contrast is purely photoelectric and lives at low energy. We simulate the whole particle – a magnetite core plus a polyacrylic-acid coating – with the coating fraction taken per formulation from the article's thermogravimetry. We simulate a C-arm angiography system with a fan-beam short scan, Parker weighting, cosine and Shepp-Logan filtering, a distance-weighted backprojector, and an always-on water beam-hardening precorrection. We compare an energy-integrating detector (EID) with a photon-counting detector (PCD), optimizing the tube voltage, aluminium filtration, and PCD energy bins for iron, and we score detectability by iron ΔHU, contrast-to-noise ratio (CNR), and the Rose criterion. We study homogeneous cellular uptake and vascular delivery, and verify in three dimensions on rabbit anatomy. The optimum is a soft 70 kVp beam with 5 mm aluminium, which gives an iron CNR of 16.9 at 1 mg Fe/ml for the photon-counting detector, while the lowest detectable iron at the Rose criterion is about 0.4 mg Fe/ml at the optimum. Overall, SPIONs sit near the CT detection limit at realistic load, and tumor cell density is the dominant factor for visibility.

Keywords: SPION, iron oxide nanoparticles, computed tomography, photon counting, detectability, contrast-to-noise ratio, spectral optimization, simulation

1. INTRODUCTION

Superparamagnetic iron-oxide nanoparticles (SPIONs) are a workhorse of nanomedicine: magnetic fields steer them for local drug delivery, they can be heated for hyperthermia, and they can label cells for tracking. The particles we study are those of Heinen et al.,[2] polyacrylic-acid (PAA)-coated magnetite (Fe_3O_4) clusters. The cores are about 8 and 12 nm, and they assemble into hydrodynamic clusters of about 70–250 nm that are used at iron concentrations of about 1–10 mg Fe/ml.

A clinical question follows: once these particles reach a target, can we simply see them in a routine CT scan? The answer matters for dose planning, for reading incidental findings, and most of all for image-guided delivery, where one wants to confirm that particles reached the target without a second imaging modality.[9]

At first sight iron looks promising, because it is denser and higher in atomic number than soft tissue. The problem is the amount, since at realistic loading the delivered iron mass is small: a few milligrams of particles spread through a cubic-centimetre tumor leave only tenths of a milligram of iron per millilitre, roughly ten times below the iodine concentrations that give comfortable CT enhancement. Whether this little iron rises above image noise is therefore not obvious.

The physics is also unhelpful. Iron has a K-edge at 7.1 keV, far below the diagnostic window, so there is no K-edge jump to exploit as there is for iodine or gadolinium. The iron contrast is purely photoelectric, which

Send correspondence to A.M.: E-mail: andreas.maier@fau.de

means it is strongest at low energy and fades quickly as energy rises. This shapes every result: a soft spectrum helps, a hard spectrum hurts, and photon counting can only help where low-energy photons survive the body.

Despite a large SPION literature and a surge of interest in photon-counting CT,[4,10,12,13] there is still little quantitative simulation of SPION CT detectability across modern detectors, and we address this gap. Our contributions are: (i) a delivered-mass dose model tied to a real magnetite SPION formulation, with a whole-particle material model and a per-formulation coating fraction from thermogravimetry; (ii) a spectrum-and-detector optimization of tube voltage, aluminium filter, and PCD energy bins for iron; (iii) two biological uptake models – homogeneous cellular uptake and vascular fresh injection – grounded in the article's measured cellular loading; and (iv) a 3D cone-beam verification on real rabbit anatomy. The full pipeline is open source at https://github.com/akmaier/SPIONvsXRay (built on the CONRAD framework), and every intermediate result is public for audit.

2. MATERIALS AND METHODS

We fix the parts in order: the nanoparticle and dose model, the phantom, the imaging chain, the two detectors, and the experiments and metrics. Each choice keeps the $c_{\mathrm{Fe}} = 0$ case identical to background, so any measured contrast is attributable to iron alone.

2.1 Nanoparticle and delivered-mass dose model

We model the whole particle, not iron alone: a magnetite (Fe_3O_4) core plus a PAA coating (monomer $C_3H_4O_2$). The iron mass fraction of magnetite is 0.724 ($3 \times 55.85/231.5$), and the bound oxygen is nearly tissue-equivalent, so it adds only a few percent to the attenuation per gram of iron. The PAA coating is low-Z (C/H/O) and essentially water-equivalent, so it is negligible for the X-ray attenuation; we still include its mass so the reported particle concentration is honest.

We keep two concentrations distinct throughout. The iron concentration c_{Fe} (mg Fe/ml) drives the attenuation, while the whole-particle concentration c_{NP} (mg SPION/ml) is derived from it through the coating fraction. From an iron concentration we recover the whole-particle concentration by

$$c_{\mathrm{NP}} = \frac{c_{\mathrm{Fe}}}{0.724\,(1-\varphi)}, \tag{1}$$

where φ is the PAA coating mass fraction. All contrast numbers are reported as mg Fe/ml in the tumor.

The coating fraction is taken per formulation from the article's supplementary thermogravimetry (TGA).[2] The inorganic residual is 87.7% for SPION I (12 nm cores) and 66.4% for SPION II (8 nm cores). On heating, the magnetite in the residual oxidizes to Fe_2O_3, a mass gain of about 3.4%, and this gain is the same for the maghemite (γ-Fe_2O_3) and hematite polymorphs, so the coating-fraction estimate is unaffected. Correcting for it, the magnetite fraction is 84.8% (SPION I) and 64.2% (SPION II), with the complement being PAA, so $\varphi \approx 0.15$ for SPION I and $\varphi \approx 0.36$ for SPION II. Each formulation is simulated with its own coating, which gives $c_{\mathrm{NP}} \approx 1.63\,c_{\mathrm{Fe}}$ (SPION I) and $c_{\mathrm{NP}} \approx 2.16\,c_{\mathrm{Fe}}$ (SPION II). Adding the coating mass moves the monochromatic iron ΔHU by only about 3.6% at $\varphi = 0.15$, so the attenuation stays iron-dominated. Table 1 lists the composition.

We use a delivered-mass dose model, not a fixed vial concentration: we spread a fixed particle mass through the tumor and ask how much iron that leaves per millilitre. We anchor at 6 mg of SPIONs delivered for the 10 mg/ml formulation over an 8 cm^3 tumor and scale with concentration, giving the linear relation

$$c_{\mathrm{Fe}} = 0.0543 \cdot c_{\mathrm{form}} \quad [\mathrm{mg\,Fe/ml}], \tag{2}$$

where c_{form} is the formulation concentration in mg/ml. The reference 6 mg dose gives $c_{\mathrm{Fe}} = 0.54$ mg Fe/ml, and even the top of the sweep is only about 1 mg Fe/ml: by design, we model the diluted delivered dose in the tumor, not the concentrated vial.

The material attenuation is taken from CONRAD's database. As a cross-check, the magnetite mass attenuation agrees with an independent NIST-based model to five decimals, so the physics underlying every result is validated. At a representative 60 keV, 1 mg Fe/ml raises the attenuation by only about 6.7 HU, which sets the scale of the study. The iron K-edge sits at 7.1 keV, far below the diagnostic range, so there is no edge to exploit. Figure 1 shows the linear attenuation of soft tissue, cortical bone, and iron-loaded tumor, and where the iron contrast lives in energy.

Table 1. Nanoparticle composition (E1). The whole particle is a magnetite core plus a PAA coating. The coating fraction φ is from the article's TGA. Iron mass fraction of magnetite is 0.724. c_{NP} per unit c_{Fe} follows Eq. (1).

Component	Description	Value
Core	Magnetite Fe_3O_4	iron fraction $F_{\mathrm{Fe}} = 0.724$
Coating	Polyacrylic acid (PAA), $(C_3H_4O_2)_n$	low-Z, tissue-equivalent
φ (SPION I)	PAA mass fraction (TGA, 12 nm core)	0.15
φ (SPION II)	PAA mass fraction (TGA, 8 nm core)	0.36
Example conversion $c_{\mathrm{Fe}} = 1.0\,\mathrm{mg\,Fe/ml} \rightarrow$ whole-particle c_{NP}:		
SPION I	$c_{\mathrm{NP}} = c_{\mathrm{Fe}}/[F_{\mathrm{Fe}}(1-\varphi)]$	1.62 mg SPION/ml
SPION II		2.16 mg SPION/ml

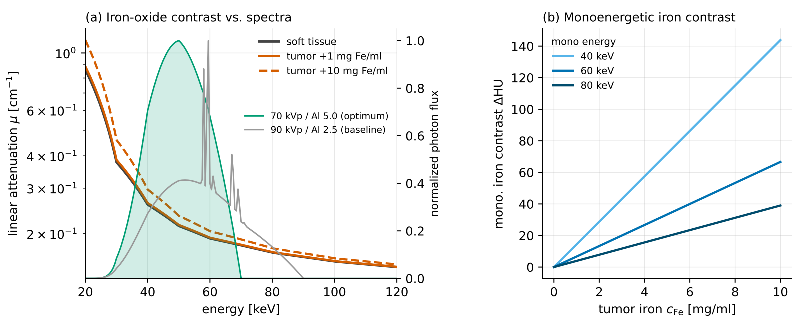

Figure 1. Physics of iron contrast (E1). Left: linear attenuation versus energy for soft tissue, cortical bone, and iron-loaded tumor. Iron has no usable K-edge (7.1 keV); the excess attenuation is photoelectric and concentrated at low energy. Right: per-energy iron contrast through the phantom, motivating a soft spectrum and low-energy PCD bins.

2.2 Digital phantom

We use a rabbit-scale, electron-density-style calibration phantom: a 110 mm soft-tissue cylinder. It holds the iron-concentration inserts on a common ring, so that beam-hardening cupping is common-mode across inserts, together with a cortical-bone rod that acts as a deliberate beam-hardening source and makes the bone factor meaningful. Each insert is a disk of iron-loaded soft tissue, with iron added to the tissue matrix by the mixture rule, so a zero-iron insert equals background exactly. Because the soft tissue is a water-equivalent body material, the zero-iron baseline stays clean. The tumor is an $8\,\mathrm{cm}^3$ sphere ($r = 12.4$ mm), and the reconstruction field of view is 20 cm. Figure 2 shows an axial slice and an example EID/PCD reconstruction.

2.3 Fan-beam imaging pipeline

The reconstruction uses established implementations from the CONRAD framework,[1,11] in its geometry convention. The fan-beam geometry uses a source-to-isocentre distance of 750 mm and a virtual detector at the isocentre sampled at $\Delta_T = 1$ mm, with 500 views over a minimal short scan (180° plus fan). The chain is forward projection, Parker weighting, cosine filter, Shepp-Logan ramp filter, and a distance-weighted fan-beam backprojector, on a 0.5 mm isotropic grid. A water beam-hardening precorrection is *always on*, calibrated so a water cylinder reconstructs flat to about 0.1%; it is not a factor of the study.

Reconstruction runs on the GPU through OpenCL, here on an Apple M1 laptop, about 850× faster than the CPU path.

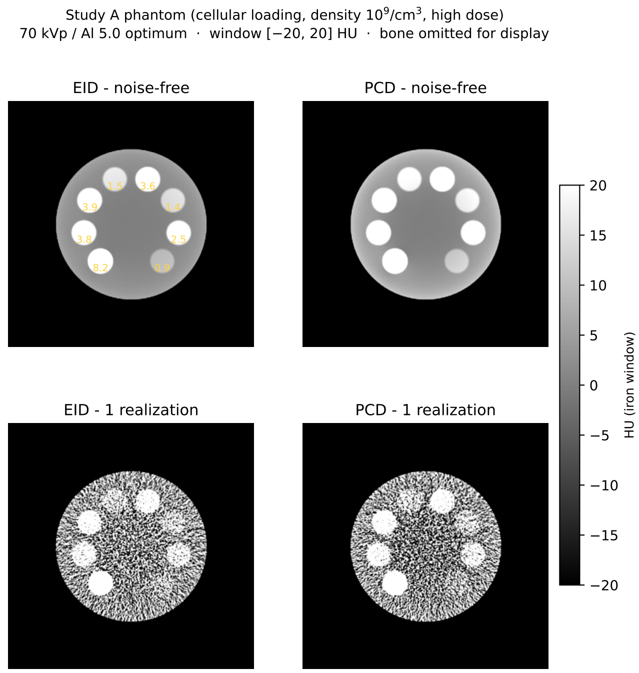

Figure 2. Digital phantom and reconstruction. Left: axial slice of the rabbit-scale phantom (soft-tissue cylinder, iron-concentration inserts on a ring, cortical-bone rod). Center/right: example EID and PCD reconstructions through the fan-beam short-scan pipeline.

2.4 Detectors: EID and PCD

Both detectors start from the same polychromatic forward model. For each energy E the transmitted photons are $N(E) = N_0\, s(E)\, e^{-\tau(E)}$, with s the normalized spectrum and τ the material path integral. Quantum noise is Poisson per energy, drawn on the GPU with a counter-based Philox random-number generator[7] and transformed-rejection Poisson sampling.[8]

The energy-integrating detector (EID) weights photons by energy, $S = \sum_E N(E)\, E$, and its variance is the exact second moment $\sum_E N(E)\, E^2$, not a single Poisson draw on the summed signal. The line integral is $p = -\log(S/S_{\mathrm{air}})$, and a single water precorrection linearizes the response.

The photon-counting detector (PCD) sorts photons into three energy bins, combined with a *post-combination matched filter*: a matched-filter weighted sum of the bin counts, then a single logarithm, then one water precorrection on the combined spectrum. The weights $w_b \propto S_b/V_b$ are the CNR-optimal weighting that reaches the ideal-observer ceiling. Because this correction uses the combined spectrum rather than each bin separately, a small residual beam hardening remains, which is visible in the noise-free reconstructions. Combining in the count domain before the logarithm is robust to the photon-starved low-energy bin, whose per-bin logarithm would diverge when its counts reach zero. This scheme gives about 1.2× the EID CNR, whereas a per-bin pre-sum correction, in which each bin is corrected on its own spectrum before the sum, is noisier and gives only about 1.0×, so we do not use it.

2.5 Experiments

We run five experiments. Table 2 lists the shared simulation parameters.

E1 – material and physics validation. We confirm the magnetite attenuation against NIST, confirm the iron K-edge is out of range, and confirm the whole-particle model keeps the attenuation iron-dominated (Fig. 1, Table 1).

E2 – spectrum and detector optimization. Iron contrast is photoelectric, so it favours low energy. We sweep the C-arm tube voltage (70–120 kVp) crossed with aluminium filtration (Al 1.0–8.0 mm) for both detectors, end-to-end. Hardening metals (Cu, Sn) only hurt iron contrast, so we drop them. E2 then (a) picks the CNR-optimal spectrum, (b) re-optimizes the PCD three-bin thresholds for it, and (c) feeds the winner to E3–E5. All later experiments run at this E2 optimum *and* at a 90 kVp baseline, for comparison against a standard C-arm spectrum.

E3 – Study A, homogeneous cellular uptake. Iron internalized by tumor cells gives a roughly uniform distribution. We sample the article's measured cellular loading (Heinen et al., Fig. 5, B16-F10 melanoma; 0/24 h): SPION I-113 nm 8.23/3.78, SPION II-115 3.86/1.52, II-98 3.60/1.37, II-76 2.51/0.85 pg Fe/cell. Each converts to a tumor iron concentration through a cell density. Cell density is uncertain, so it is a factor with three levels (10^8, 3×10^8, 10^9 cells/cm^3), giving a detectability band. Each configuration uses its formulation's coating; we also cross the bone and dose factors. Table 3 lists the loading-by-density grid.

E4 – Study B, vascular / fresh injection. Freshly injected SPIONs are still in the blood carrier inside the vessels, before cellular uptake. The contrast is confined to 150 μm vessels that occupy about 10% of the tumor volume at a tenfold local concentration, and the vessel-local level is the independent variable (5–15 mg Fe/ml). Because the 0.5 mm voxel cannot resolve the vessels, each insert is homogenized to its tumor-mean iron. We then add the second-order effects the homogeneous model omits: the beam-hardening nonlinearity from concentrating iron, and the structural noise from the binomial per-voxel vessel count.

E5 – three-dimensional cone-beam verification. We verify the result in three dimensions on real anatomy. We use the post-mortem RabbitCT rabbit volume[5] and its C-arm geometry, decoded from the benchmark's projection matrices (745 mm source-to-isocentre, 1196 mm source-to-detector, 496 views over 198°). We forward-project poly-energetically, insert a material-separated 8 cm^3 SPION tumor, and reconstruct with cone-beam FDK. E5 also runs at the E2 optimum and the 90 kVp baseline.

2.6 Metrics

We score detectability with three quantities. The iron contrast ΔHU is the mean insert value minus a local annular background, corrected by subtracting the zero-iron insert so only iron remains. The CNR is that contrast divided by the background noise over noise realizations. An insert is detectable when its CNR clears the Rose criterion;[3,6] we report both CNR ≥ 3 and ≥ 5, from 30 noise realizations per cell. The CNR is a simple surrogate for formal model-observer detectability on the reconstructed image.[14]

Table 2. Shared simulation parameters. The E2 optimum and a 90 kVp baseline are both run in E3–E5.

Parameter	Value
Spectrum	E2 optimum (70 kVp, Al 5 mm) and 90 kVp baseline, W anode
Photons per pixel	low 20,000 / high 100,000 (Poisson), factor
Geometry (2D)	C-arm fan beam, 500 views, minimal short scan, Parker weighted
Source-to-isocentre / detector	750 mm / 1200 mm, $\Delta_T = 1$ mm
Reconstruction (2D)	fan FBP (cosine + Shepp-Logan + distance backprojector), 0.5 mm voxels
Beam hardening	water precorrection always on
Phantom	110 mm soft-tissue cylinder + cortical-bone rod (factor)
Tumor	8 cm^3 ($r = 12.4$ mm), iron-loaded
Detectors	EID; 3-bin PCD (post-combination matched filter)
3D (E5)	RabbitCT geometry (745/1196 mm, 496 views, 198°), cone-beam FDK
Noise realizations	30 per cell

3. RESULTS

3.1 E2 spectrum and detector optimum

Because iron contrast is photoelectric, it should favour a soft spectrum, and the E2 sweep confirms this: CNR falls as tube voltage rises and as aluminium hardens the beam. The CNR-optimal spectrum is a soft 70 kVp beam with 5 mm aluminium, and at this optimum, at the representative 1 mg Fe/ml, the EID reaches a CNR of

9.6 and the PCD 16.9 (1.20×), with the matched-filter PCD reaching 98% of the ideal-observer ceiling. For this spectrum the PCD bins re-optimize to edges of 10.0/37.5/47.5/70.0 keV, and the optimum lifts the EID CNR by about 24% over the 90 kVp baseline (7.8 → 9.6). Figure 3 shows the CNR surface with the optimum marked, and Table 6 reports the EID versus PCD CNR.

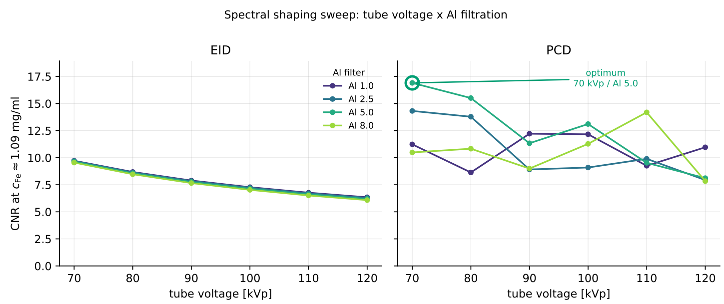

Figure 3. E2 spectral optimization. CNR versus tube voltage across aluminium filters, for EID and PCD. Lower tube voltage helps; hardening filters hurt. The selected CNR-optimal spectrum (70 kVp, Al 5 mm, PCD bin edges 10/37.5/47.5/70 keV) feeds E3–E5.

3.2 Study A: homogeneous cellular uptake and the cell-density band

The central result is that SPIONs sit near the CT detection limit at realistic load. Figure 4 plots iron ΔHU and CNR against tumor iron concentration for both detectors, with the Rose lines and a shaded cell-density band. The band comes from the three cell-density levels: the same cellular loading maps to different tumor iron concentrations as density changes, so visibility is dominated by how densely the tumor is packed. The fresh SPION I loading (8.23 pg Fe/cell) gives tumor concentrations of 0.82/2.47/8.23 mg Fe/ml at the three densities, and at high dose it clears the Rose CNR = 5 criterion already at 10^8 cells/cm^3, while at low dose it clears it only from 3×10^8 upward, so detection is dose-limited rather than detector-limited. The lowest detectable iron at Rose ≥ 5 is 0.41 mg Fe/ml for the PCD at the optimum (3×10^8, high dose) against 0.75 mg Fe/ml for the EID. At the reference load (0.54 mg Fe/ml) the CNR sits on the Rose boundary, while at the top of the band the PCD CNR peaks near 96. Figure 5 shows a reconstruction montage across the density levels, and Table 3 lists the loading-by-density grid and Table 4 the detection thresholds.

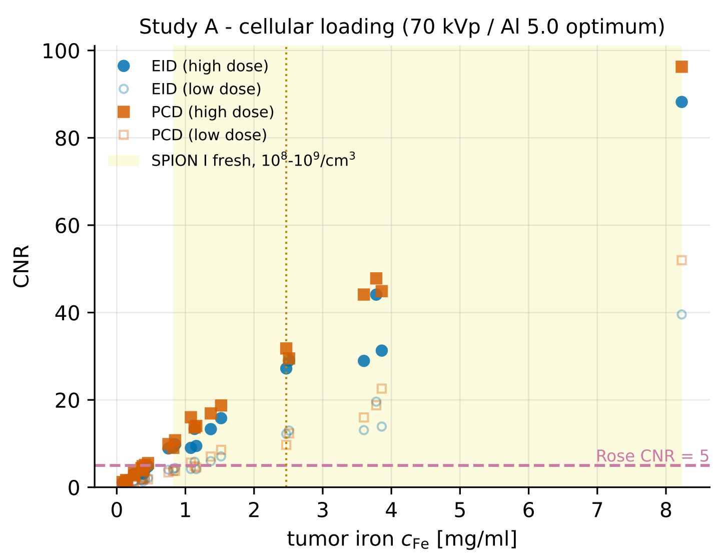

Figure 4. Study A detectability. Iron ΔHU (left) and CNR (right) versus in-tumor iron concentration for EID and PCD, with the Rose 3 and 5 lines. The shaded band spans the three tumor cell densities (10^8–10^9 cells/cm^3); it shows that visibility is dominated by cell density.

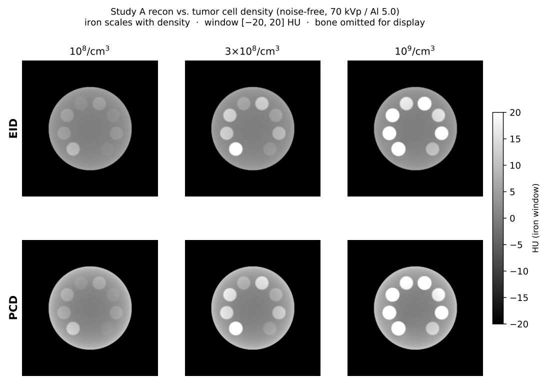

Figure 5. Study A cell-density montage. Reconstructions of the iron-loaded tumor at increasing tumor cell density ($10^8 \rightarrow 3 \times 10^8 \rightarrow 10^9$ cells/cm^3), at fixed cellular loading.

Table 3. Study A loading × cell-density grid. Measured cellular loading (Heinen et al., Fig. 5; 0 h / 24 h) converted to tumor iron concentration c_{Fe} [mg Fe/ml] via cell density ρ.

			c_{Fe} [mg/ml] at cell density		
Configuration	Formulation	pg Fe/cell	10^8	3×10^8	10^9
I_113 (0 h)	I	8.23	0.82	2.47	8.23
I_113 (24 h)	I	3.78	0.38	1.13	3.78
II_115 (0 h)	II	3.86	0.39	1.16	3.86
II_115 (24 h)	II	1.52	0.15	0.46	1.52
II_98 (0 h)	II	3.60	0.36	1.08	3.60
II_98 (24 h)	II	1.37	0.14	0.41	1.37
II_76 (0 h)	II	2.51	0.25	0.75	2.51
II_76 (24 h)	II	0.85	0.09	0.26	0.85

Table 4. Detection thresholds – the lowest c_{Fe} [mg Fe/ml] reaching the Rose CNR ≥ 5 criterion (bone off) – per detector, cell density, dose, and spectrum. A dash means the criterion is not reached within the swept range.

		Optimum (70 kVp/Al5)		Baseline (90 kVp/Al2.5)	
Detector	Cell density	low dose	high dose	low dose	high dose
EID	10^8	–	0.82	–	0.82
EID	3×10^8	1.13	0.75	2.47	0.75
EID	10^9	1.37	0.85	1.52	0.85
PCD	10^8	–	0.82	–	0.82
PCD	3×10^8	1.08	0.41	1.13	0.75
PCD	10^9	1.37	0.85	1.37	0.85

3.3 Bone off versus bone on

The cortical-bone rod is a deliberate beam-hardening source, and we repeat Study A and Study B with the rod present. Bone raises the iron ΔHU a little, because the hardened beam shifts the local response, but it raises the local noise more, through residual streaks that the local-background and zero-iron subtraction suppress but do not remove, so the net effect is a modest CNR drop. In Study A at the optimum, high dose, SPION I fresh, the EID CNR falls from 27.2 to 18.4 at 3×10^8 cells/cm^3 ($c_{\mathrm{Fe}} = 2.47$) and from 88.2 to 54.9 at 10^9 ($c_{\mathrm{Fe}} = 8.23$); the PCD falls from 31.8 to 21.5 and from 96.3 to 86.3. Study B behaves the same: at the optimum, high dose, the EID CNR falls from 6.1 to 3.5 at a 5 mg Fe/ml vessel level and from 13.3 to 9.1 at 15 mg Fe/ml. The Rose

≥ 5 thresholds stay in the 0.4–1.5 mg Fe/ml regime, so iron detectability is not qualitatively changed by bone. Table 5 lists the bone off versus on CNR for representative cells.

Table 5. Bone off versus bone on. Iron CNR at matched load, for representative Study A and Study B cells at the E2 optimum, high dose. Bone raises ΔHU slightly via beam hardening but raises local noise more, so CNR drops; the detection thresholds stay in the 0.4–1.5 mg Fe/ml regime.

Cell	Config	c_{Fe} [mg/ml]	CNR bone off	CNR bone on
Study A, optimum, high dose, SPION I fresh				
EID	3×10^8	2.47	27.2	18.4
EID	10^9	8.23	88.2	54.9
PCD	3×10^8	2.47	31.8	21.5
PCD	10^9	8.23	96.3	86.3
Study B, optimum, high dose, vessel-local level				
EID	5 mg/ml	0.50	6.1	3.5
EID	15 mg/ml	1.50	13.3	9.1
PCD	15 mg/ml	1.50	15.7	9.8

3.4 Study B: vascular / fresh injection

Real fresh-injection uptake is vascular, not uniform, but the decisive fact is one of scale. A 0.5 mm voxel is more than three times wider than a 150 μm vessel, so each voxel partial-volume-averages many vessels. With mass conserved, the volume-averaged iron per voxel is unchanged, and the tumor-ROI contrast matches the homogeneous case. The second-order effects are small: the beam-hardening nonlinearity is well below 10^{-3} HU, and the binomial per-voxel vessel count adds texture that averages out over the ROI, so the ROI-mean detectability tracks the total delivered iron, as in Study A. At high dose the ROI-mean CNR crosses the Rose 5 line at a vessel-local level of 5 mg Fe/ml (EID) and 7 mg Fe/ml (PCD); at low dose at 13 (EID) and 11 mg Fe/ml (PCD). Figure 6 shows that vascular uptake is CT-indistinguishable from homogeneous uptake at realistic resolution.

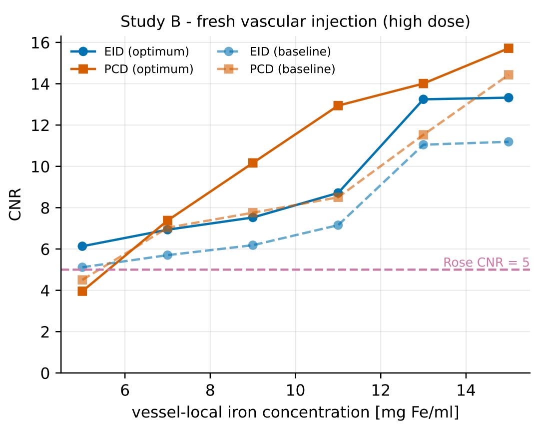

Figure 6. Study B, vascular uptake. The ROI-mean iron concentration is invariant (mass conservation), while the per-voxel peak approaches the resolved tenfold value only as the voxel shrinks toward the 150 μm vessel size. At the 0.5 mm CT voxel the vessels are unresolved.

3.5 EID versus PCD

Photon counting helps, but only partly. The post-combination matched-filter PCD gives about 1.2× the EID CNR when the low-energy bin carries enough counts, but at realistic dose that bin is photon-starved through the body, so the gain shrinks. Averaged over the Study A grid (bone off), the PCD CNR is about 1.15× the EID at the optimum and 1.28× at the baseline (mean ≈ 1.2×), because the softer optimum lifts the EID more

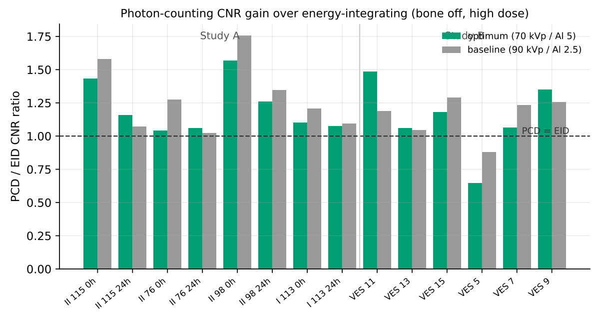

Figure 7. EID versus PCD. Per-energy iron detectability through the phantom, with the re-optimized three-bin PCD thresholds (10/37.5/47.5/70 keV). The contrast lives where transmitted photons are scarce, which limits the realized PCD gain to about 1.2× EID.

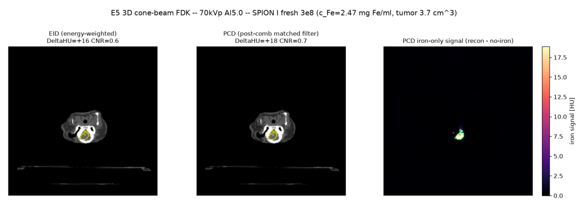

Figure 8. E5 three-dimensional verification in real anatomy. A cervical slice of the post-mortem RabbitCT rabbit, reconstructed with cone-beam FDK at the E2 optimum, with an inserted SPION I fresh tumor ($c_{Fe} = 2.47\,\mathrm{mg\,Fe/ml}$). Left: EID (+16 HU). Middle: PCD, post-combination matched filter (+18 HU). Right: the PCD iron-only signal (reconstruction minus a no-iron reference), which localizes the tumor while the anatomy cancels.

Table 6. EID versus PCD CNR by cellular-loading configuration, at cell density 10^9 cells/cm^3, high dose, bone off, for the E2 optimum and the 90 kVp baseline (post-combination matched filter).

	Optimum (70 kVp/Al5)			Baseline (90 kVp/Al2.5)	
Configuration	EID	PCD	PCD/EID	EID	PCD
I 113 0h	88.2	96.3	1.09	72.1	92.6
I 113 24h	44.1	47.8	1.08	35.8	38.5
II 115 0h	31.3	44.9	1.43	25.9	42.5
II 115 24h	15.8	18.7	1.18	13.0	13.1
II 98 0h	28.9	44.1	1.53	24.0	42.8
II 98 24h	13.3	16.9	1.27	11.0	14.5
II 76 0h	29.0	29.5	1.02	24.0	32.1
II 76 24h	9.8	10.8	1.10	8.1	8.4

and narrows the relative gap. The per-bin pre-sum variant gives essentially no gain ($\sim 1.0\times$) and is noisier, so it is not used. Figure 7 shows where the iron contrast lives in energy and the re-optimized bin thresholds, and the gain sits in a band that receives few transmitted photons.

3.6 Three-dimensional verification

Finally we verify the result in real anatomy. We segment the post-mortem RabbitCT volume into soft tissue, adipose, and bone, add an iron-loaded neck tumor, forward-project each material in 3D cone-beam geometry at the E2 optimum, and reconstruct with cone-beam FDK for both detectors. The iron signal is positive and scales linearly at about 6.6 HU per mg Fe/ml, matching 2D. The PCD gives a higher CNR at every level, a gain of about 1.14×, as E2 predicts. For SPION I fresh at a mid cell density (3×10^8, $c_{Fe} = 2.47$) the tumor reads $+16/+18$ HU (EID/PCD); at the highest load (10^9) $+55/+60$ HU. The 3D result agrees with the 2D detection limit (Fig. 8).

4. DISCUSSION

The picture is coherent. At realistic load, SPIONs deposit little iron, which produces only a few Hounsfield units of contrast, and the CNR lands near the Rose criterion, not comfortably above it. The reference dose is therefore borderline-visible: detectable in principle at low noise and a soft spectrum, but not a robust everyday finding.

Which factor helps most? Cell density: the same cellular loading maps to a wide range of tumor iron concentrations depending on how densely the tumor is packed, so visibility is dominated by cell density, not the formulation alone. The next lever is the spectrum, because a soft beam raises the photoelectric iron contrast, while photon counting gives only a modest $1.2\times$ gain, and only when the low-energy bin survives the body. Beam-hardening correction does not change iron detectability, because iron contrast is a genuine material signal, not a cupping artifact.

The study has clear limitations. It is a simulation with idealized, ideal-energy detectors and no scatter or patient motion. The 2D phantom is a round, rabbit-scale cylinder rather than full anatomy, and soft tissue is a water-equivalent proxy. Study B shows sub-resolution vessels do not change detectability at the same total dose, and E5 brings in real anatomy, but a full 3D detectability sweep is future work. The open decisions – tumor cell density (Study A) and injection concentration (Study B) – are reported as bands.

5. CONCLUSION

Can superparamagnetic iron-oxide nanoparticles be seen in CT? At realistic load, only barely: our simulation places the detection limit near the reference SPION dose, at a few Hounsfield units and a CNR on the Rose boundary. Tumor cell density is the dominant factor, while photon counting adds about $1.2\times$ and a soft spectrum is the most effective lever. Routine CT confirmation of SPION delivery is therefore feasible only at the upper end of realistic loading, with a spectrum tuned for iron, and full 3D anatomical detectability and experimental validation are the next steps.

Code and data availability

The complete pipeline, intermediate results, and figure sources are public (MIT-licensed), with an audit dashboard at `https://akmaier.github.io/SPIONvsXRay/`. The GPU OpenCL detector, noise generator, analytic fan projector, water precorrection tool, and back-projector are available in CONRAD.

REFERENCES

[1] A. Maier et al., "CONRAD – a software framework for cone-beam imaging in radiology," *Medical Physics* **40**(11), 111914 (2013). doi:10.1118/1.4824926

[2] L. Heinen et al., "Ultrasound-assisted size tuning of polyacrylic acid coated magnetic nanoparticle clusters for biomedical applications," *Ultrasonics Sonochemistry* **130**, 107876 (2026). doi:10.1016/j.ultsonch.2026.107876

[3] A. Rose, *Vision: Human and Electronic*, Plenum Press, New York (1973). ISBN 978-0-306-30732-2

[4] M. J. Willemink et al., "Photon-counting CT: technical principles and clinical prospects," *Radiology* **289**(2), 293–312 (2018). doi:10.1148/radiol.2018172656

[5] C. Rohkohl, B. Keck, H. G. Hofmann, and J. Hornegger, "Technical Note: RabbitCT – an open platform for benchmarking 3D cone-beam reconstruction algorithms," *Medical Physics* **36**(9), 3940–3944 (2009). doi:10.1118/1.3180956

[6] M. J. Tapiovaara and R. F. Wagner, "SNR and DQE analysis of broad spectrum X-ray imaging," *Phys. Med. Biol.* **30**(6), 519–529 (1985). doi:10.1088/0031-9155/30/6/002

[7] J. K. Salmon et al., "Parallel random numbers: as easy as 1,2,3," in *Proc. SC'11*, ACM (2011). doi:10.1145/2063384.2063405

[8] W. Hörmann, "The transformed rejection method for generating Poisson random variables," *Insurance: Mathematics and Economics* **12**(1), 39–45 (1993). doi:10.1016/0167-6687(93)90997-4

[9] A. Maier et al., "A gentle introduction to deep learning in medical image processing," *Z. Med. Phys.* **29**(2), 86–101 (2019). doi:10.1016/j.zemedi.2018.12.003

[10] F. Goldmann et al., "Challenging Hounsfield Unit cutoffs: spectral thresholding for synthetic coronary plaque phantoms on photon-counting CT," *J. Med. Imaging* **13**(2), 024003 (2026). doi:10.1117/1.JMI.13.2.024003

[11] M. T. Manhart et al., "Dynamic Iterative Reconstruction for Interventional 4-D C-Arm CT Perfusion Imaging," *IEEE Trans. Med. Imaging* **32**(7), 1336–1348 (2013). doi:10.1109/TMI.2013.2257178

[12] S. Faby, S. Kuchenbecker, S. Sawall, D. Simons, H.-P. Schlemmer, M. Lell, and M. Kachelrieß, "Performance of today's dual energy CT and future multi energy CT in virtual non-contrast imaging and in iodine quantification: A simulation study," *Med. Phys.* **42**(7), 4349–4366 (2015). doi:10.1118/1.4922654

[13] B. De Man et al., "CatSim: a new computer assisted tomography simulation environment," *Proc. SPIE* **6510**, 65102G (2007). doi:10.1117/12.710713

[14] A. Wunderlich and F. Noo, "Image covariance and lesion detectability in direct fan-beam x-ray computed tomography," *Phys. Med. Biol.* **53**(10), 2471–2493 (2008). doi:10.1088/0031-9155/53/10/002